

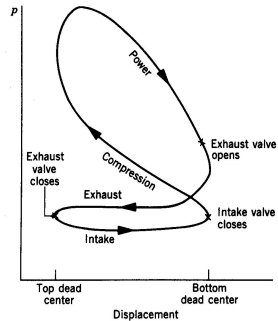
Actual Cycles

Comparison

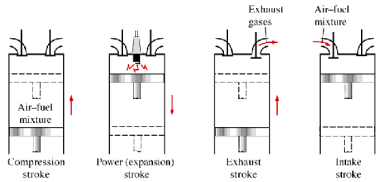
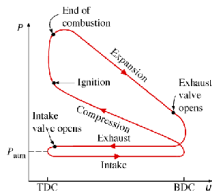
- 1 Working substance, change in chemical composition
- 2 Variation of specific heat
- 3 Dissociation of combustion products
- 4 Progressive combustion, Incomplete combustion
- 5 Heat transfer
- 6 Exhaust blowdown loss
- 7 Gas leakage, friction

Actual cycle

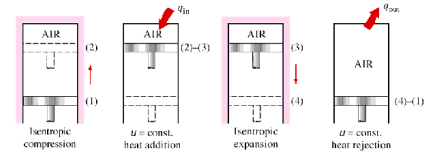
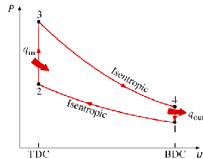
Otto cycle



Actual Working of Otto Cycle



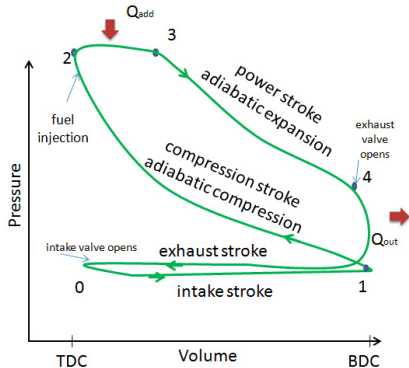
(a) Actual four-stroke spark ignition engine



(b) Ideal Otto cycle

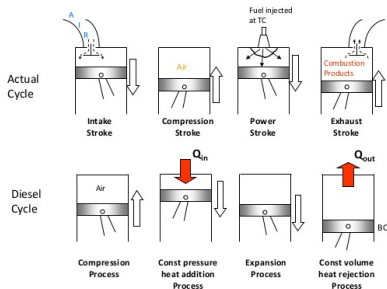
a) Actual Otto Cycle working b) Ideal Otto Cycle working

Actual Working of Diesel Cycle



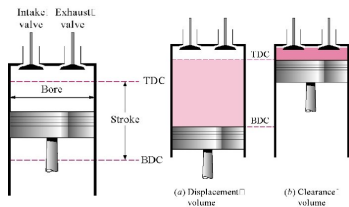
Actual Diesel Cycle working

Early CI Engine Cycle and the Thermodynamic Diesel Cycle

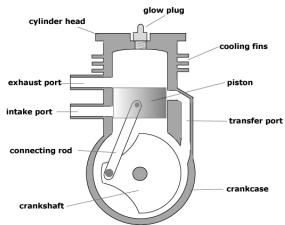


valves and ports

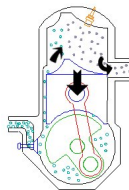
4 stroke engine valves



2 stroke engine

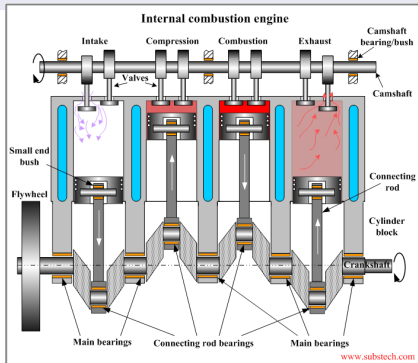


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valves and ports

4 stroke engine valves

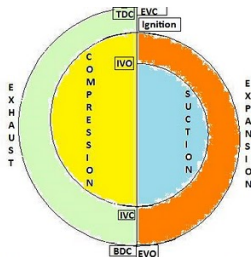


Valves



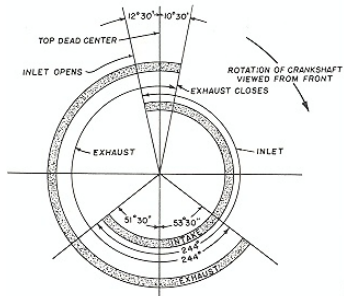
Valve Timing

Theoretical valve timing



Valve Timing Diagram-4stroke engine/otto cycle (Theoretical)

Actual valve timing



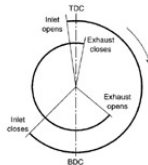
Actual Valve Timing

Actual valve timing diagram

Comparison

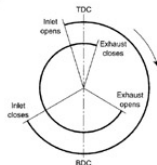
- 1 Engine Capacity
- 2 Manufacturer Design
- 3 Engine parameters

(a)



Swept volume 848 cm³
 Compression ratio 8.3:1
 Max. torque – 62 Nm at 2500 rev/min
 Max. power – 31.3 kW at 4500 rev/min

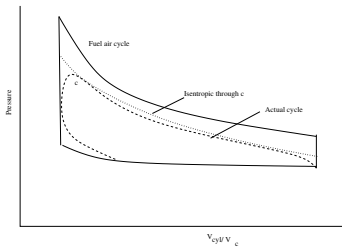
(b)



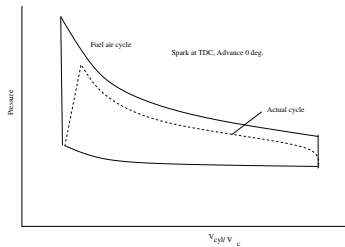
Swept volume 1955 cm³
 Compression ratio 9.2:1
 Max. torque – 151 Nm at 3500 rev/min
 Max. power – 72 kW at 5200 rev/min

Effect of advancing spark timing

Effect of time loss

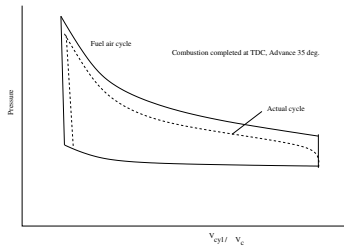


Advance 0 °

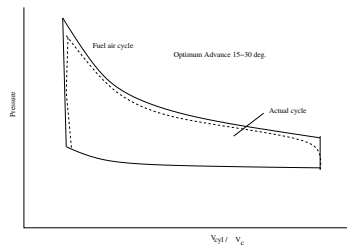


Effect of advancing spark timing

Advance by 35°



Advance optimum $15^\circ - 30^\circ$



Cycle performance for various ignition timings

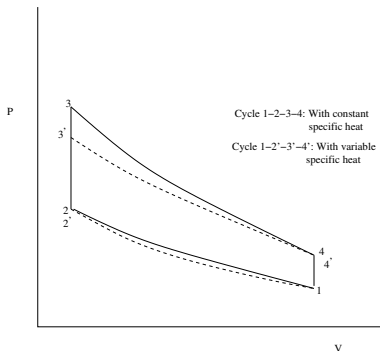
Cycle	Ignition advance	Max cycle pressure, bar	m.e.p bar	Efficiency %	Actual η / Fuel cycle η
Fuel -air cycle	0°	44	10.2	32.2	1.0
Actual	0°	23	7.5	24.1	0.75
	17°	34	8.35	26.3	0.81
	35°	41	7.6	23.9	0.74

c_p variation

Effect of c_p variation

- 1 c_p increases for most of the gases with temperature
- 2 $c_p = 1.005 \text{ kJ/kgK}$ at 300K, $c_v = 0.717 \text{ kJ/kgK}$ at 300K
- 3 $c_p = 1.343 \text{ kJ/kgK}$ at 2000K, $c_v = 1.055 \text{ kJ/kgK}$ at 2000K
- 4 γ decreases with increase in temperature
- 5 Loss of work during expansion

Loss of power due to c_p variation



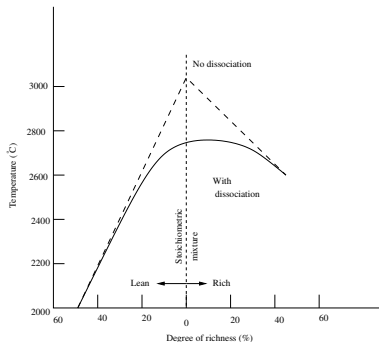
Dissociation

Effect of dissociation

- 1 disintegration of combustion products at high temperature
- 2 at 1000°C ,

$$2\text{CO}_2 + \text{Heat} \rightleftharpoons 2\text{CO} + \text{O}_2$$
- 3 at 1300°C , $\text{H}_2\text{O} + \text{Heat} \rightleftharpoons 2\text{H}_2 + \text{O}_2$
- 4 Dissociation less with rich and lean mixture

Effect of dissociation on temperature

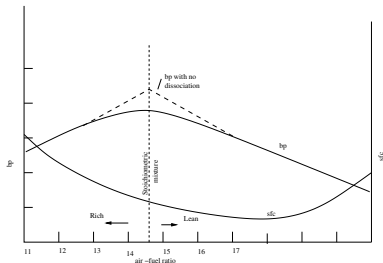


Dissociation

Effect of dissociation on power

- 1 peak temperature reduced
- 2 heat lost through exhaust
- 3 power reduced

Effect of dissociation on temperature



Engine Performance

1 Brake power (kW), $bp = \frac{2\pi NT}{60000}$, $T = (W - s)R$

2 Indicated power (kW) , $ip = \frac{LAnKp_{im}}{60 \times 1000}$

3 Indicated thermal efficiency $\eta_{ith} = \frac{ip}{\text{energy-input/s}}$

$$\eta_{ith} = \frac{ip}{m_f \times C.V}$$

4 Brake thermal efficiency $\eta_{bth} = \frac{bp}{m_f \times C.V}$

5 Mechanical efficiency $\eta_m = \frac{bp}{bp + fp}$

6 Volumetric efficiency

$$\eta_v = \frac{\text{mass-of-charge-actually-inducted}}{\text{mass-of-charge-in-swept-volume-at-ambient}}$$

7 Relative efficiency $\eta_{rel} = \frac{\text{Actual-thermal-efficiency}}{\text{Air-standard-efficiency}}$

8 Mean effective pressure $p_{im} = \frac{60 \times 1000 \times ip}{LAnK}$

9 specific fuel consumption $sfc = \frac{\text{Fuel-consumption-per-unit-time}}{\text{Power}}$

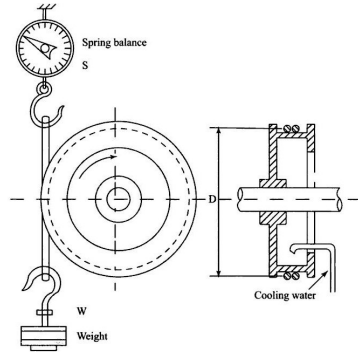
Engine Testing

Dynamometers

- 1 Rope brake dynamometer
- 2 Electric dynamometer
- 3 Hydraulic dynamometer

- 4 Brake power (kW),

$$bp = \frac{2\pi NT}{60000}, T = (W - s)R$$



Engine Performance

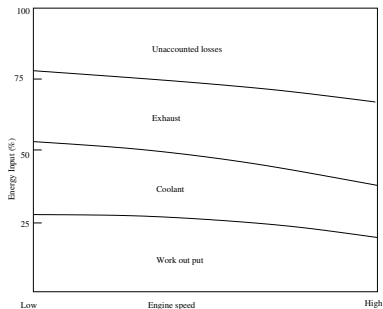
- 1 Air fuel ratio = $\frac{\text{mass-of-air}}{\text{mass-of-fuel}}$
- 2 Equivalence ratio $\phi = \frac{\text{Actual-fuel-air-ratio}}{\text{Stoichiometric-fuel-air-ratio}}$,
 $\phi=1$ - Stoichiometric-fuel-air-ratio,
 $\phi<1$ -lean mixture, $\phi>1$ -rich mixture.
- 3 Mean effective pressure $p_{bm} = \frac{60 \times 1000 \times b p_{im}}{L A n K}$
 p_{im} = indicated power, p_{im} = indicated mean effective pressure (N/m^2), L = stroke (m), A = bore area (m^2), N = Engine speed (rpm), n = number of power stroke, = N/2 for 4 stroke and N for 2-stroke, K = number of cylinder

Heat Balance

Distribution

- 1 Work output
- 2 Coolant losses
- 3 Exhaust losses
- 4 Unaccounted losses

Heat balance diagram for a SI Engine

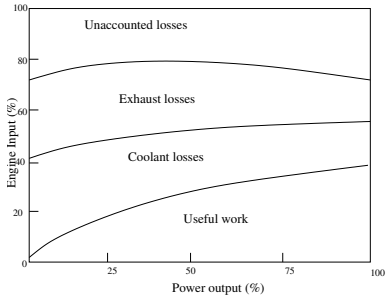


Engine Testing

Performance

- 1 Load test
- 2 Heat balance test
- 3 Morse Test
- 4 Retardation test

Heat balance diagram for a CI Engine



Frictional Power

Morse Test

- 1 Suitable for multiple cylinder engines
- 2 Suitable for SI and CI Engine
- 3 Isolate one cylinder, adjust load to bring rated speed

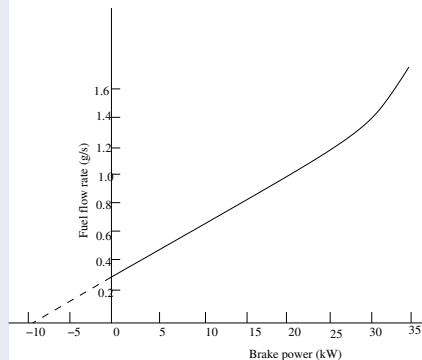
$$4 \quad bp_k = \sum_1^k ip_k - fp$$

$$5 \quad bp_{k-1} = \sum_2^k ip_k - fp$$

$$6 \quad ip_1 = bp_k - bp_{k-1}$$

$$7 \quad ip_k = \sum_1^k ip_k$$

Willan's line method



Retardation test

Test

- 1 Retarding Engine by cutting fuel
- 2 Plot speed vs time
- 3 Repeat test for no-load and 50% load

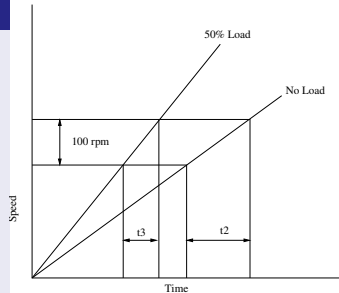
$$4 \quad T = I \frac{d\omega}{dt}, \quad T = MK^2 \frac{d\omega}{dt}$$

$$5 \quad \int_{\omega_1}^{\omega_2} d\omega = \frac{T}{MK^2} \int_{t_1}^{t_2} dt$$

$$6 \quad \text{no load, } \omega_0 - \omega_1 = \frac{T_f t_2}{MK^2}$$

$$7 \quad 50\% \text{ load, } \omega_0 - \omega_1 = \frac{(T_f + T_l) t_3}{MK^2}$$

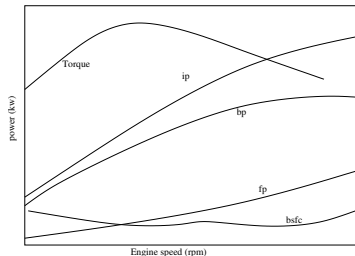
$$8 \quad \frac{t_2}{t_3} = \frac{T_l + T_f}{T_f} \Rightarrow T_f = T_l \left(\frac{t_3}{t_2 - t_3} \right)$$



Engine performance characteristics

- 1 Speed range - maximum charge inducted-maximum torque
- 2 Power output increases with speed
- 3 Air consumption increases with speed
- 4 fp increases with engine speed
- 5 bsfc drops as speed increased, levels off then increase
- 6 At low speed heat loss high, so poor efficiency, at high speed fp increases

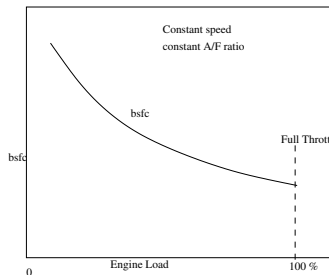
Typical SI Engine performance



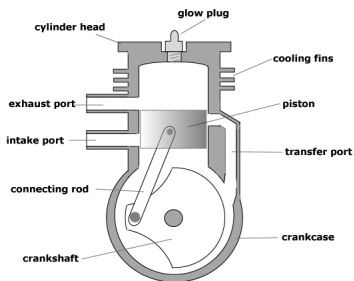
Variables Affecting Engine Performance

- 1 Combustion rate and spark timing, maximum pressure occur beginning of power stroke
- 2 Air fuel ratio to be kept most economic proportions during normal cruising speed
- 3 Higher compression ratio increases thermal efficiency, limited by engine knock
- 4 Low speed $\Rightarrow$ high heat loss, high speed $\Rightarrow$ higher fp
- 5 Mass of inducted charge- maximum possible
- 6 Heat loss - to be minimised

Typical SI Engine performance constant speed



Two stroke engines



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